

PRIME POLYNOMIALS IN LEGENDRE INTERVALS OVER $\mathbb{F}_{2^k}[t]$, II: EXACT VARIANCES VIA QUADRATIC TOWERS AND SUPERSINGULAR ARTIN–SCHREIER CURVES

RUQING CHEN

ABSTRACT. Let $q = 2^k$ and let $\mathcal{I}_f = \{f^2 + s : \deg s \leq d\}$ be the Legendre interval of a monic $f \in \mathbb{F}_q[t]$ of degree d . Chen’s Open Problem 8.2 asks why the variance of the prime-power count $\Psi(f) = \sum_{P \in \mathcal{I}_f} \Lambda(P)$ deviates by a factor as large as $q^2/3$ from the Keating–Rudnick prediction. We resolve the case $d = 4$ completely and unconditionally: using the quadratic tower $\mathbb{F}_q \subset \mathbb{F}_{q^2} \subset \mathbb{F}_{q^4} \subset \mathbb{F}_{q^8}$ we obtain a closed formula for Ψ on every class of the Hayes group $\mathcal{G}_3$, from which $\text{Var}_{\mathcal{G}^2}(\Psi)/q^5 = q(q-1)$, $\text{Var}_{\mathcal{G}}(\Psi)/q^5 = 3 - 3/q$, the full structure of the quadratic-twist correlations, and an elementary proof of Carlitz’s cubic evaluation over $\mathbb{F}_{2^{8k}}$ follow. For $d = 6$ we prove that the Legendre profile takes exactly three values governed by $[c_2 = 0]$ and $\text{Tr}(c_4/c_2^2)$, reducing the variance to three exponential sums A_6, X, Y on $U = \{\alpha \in \mathbb{F}_{q^{12}} : \text{Tr } \alpha = \text{Tr } \alpha^3 = \text{Tr } \alpha^5 = 0\}$. We show that $A_6 - q^9$ and X are sums of Frobenius traces of supersingular Artin–Schreier curves $y^2 + y = \lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x$ and reduce them to $\mathbb{F}_2$ -quadratic forms whose radicals are kernels of explicit linearised polynomials; their conjectured closed forms $A_6 = q^9 + (-1)^{k+1}(q-1)q^5$, $X = (-1)^{k+1}q^6 + q^5$ are verified exactly for $q \leq 128$ by an algorithm independent of the explicit formula, and the pure cubic part is proved for all k . The third constant $Y = q^3 m_k$ is a Frobenius trace on the nine-dimensional variety U which is not a character sum; we discuss its weight and rank. Consequently the Legendre variance at $d = 6$ obeys $\text{Var}_{\mathcal{G}^2}/q^7 \asymp q^2$, not q^4 .

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1. INTRODUCTION

Let $q = 2^k$, $d \geq 2$, $n = 2d$ and $\ell = d - 1$. For a monic $f \in \mathbb{F}_q[t]$ of degree d the *Legendre interval* is

$$\mathcal{I}_f = \{f^2 + s : s \in \mathbb{F}_q[t], \deg s \leq d\}, \quad |\mathcal{I}_f| = q^{d+1},$$

and $\Psi(f) = \sum_{P \in \mathcal{I}_f} \Lambda(P)$ counts prime powers in it with the von Mangoldt weight. In odd characteristic the sets $\mathcal{I}_f$ are ordinary short intervals and the variance of Ψ is governed by the Keating–Rudnick theorem [8], which predicts $\text{Var}(\Psi) \sim (d-2)q^{d+1}$ as $q \rightarrow \infty$. In characteristic 2 the squaring map on the Hayes group is a homomorphism, the Legendre intervals are the fibres over the subgroup of squares, and Chen [2] observed (Table 4 of *op. cit.*) that the variance over Legendre intervals exceeds the full-group variance by factors $4/3$, $16/3$, $64/3$ for $q = 2, 4, 8$ at $d = 4$ and asked for an explanation (Open Problem 8.2).

This paper is the second part of our study. Part I established the dictionary between Legendre intervals and quadratic characters of the Hayes group and verified, by exact computation for $q \leq 8$, the identities $\text{Var}_{\mathcal{G}^2}(\Psi)/q^5 = q(q-1)$ and $\text{Var}_{\mathcal{G}}(\Psi)/q^5 = 3 - 3/q$. Here we prove them for all $q = 2^k$, and we carry the analysis to $d = 6$, where a genuinely new phenomenon appears: two of the three constants that control the Legendre variance are elementary while the third is a Frobenius trace that is not a character sum.

Main results. All variances below are second moments about the global mean q^{d+1} (this is the quantity tabulated in [2]); see Section 2.

- $d = 4$ (**Section 3**). Theorem 3.2 gives Ψ on every class of the Hayes group $\mathcal{G}_3$ in closed form. Consequently (Theorems 3.4, 3.11, 3.12) for every $q = 2^k$

$$\frac{\text{Var}_{\mathcal{G}^2}(\Psi)}{q^5} = q(q-1), \quad \frac{\text{Var}_{\mathcal{G}}(\Psi)}{q^5} = 3 - \frac{3}{q}, \quad \frac{\text{Var}_{\mathcal{G}^2}}{\text{Var}_{\mathcal{G}}} = \frac{q^2}{3},$$

the deviation $\Psi - q^5$ is supported on the subgroup $\{c_1 = 0\}$, the identity class carries the deficit $-(q^4 - q^3)$, and the quadratic-twist correlations $\rho(\varepsilon)$ equal 1 on the $q-1$ characters of conductor u^2 and $(q-3)/(3(q-1))$ on all others when k is odd.

- $d = 6$, **structure (Section 4)**. Theorem 4.2: on the q^2 Legendre classes $(0, c_2, 0, c_4, 0)$ the function Ψ takes exactly three values,

$$\Psi(0, c_2, 0, c_4, 0) = q^7 + \frac{A_6 - q^9 - X}{q^2} + \frac{X}{q}[c_2 = 0] + \frac{Y}{q}[c_2 \neq 0] \chi\left(\frac{c_4}{c_2^2}\right),$$

where $A_6 = |U|$, $X = \sum_U \chi(e_2)$, $Y = \sum_U \chi(e_4)$ and $U = \{\alpha \in \mathbb{F}_{q^{12}} : \text{Tr } \alpha = \text{Tr } \alpha^3 = \text{Tr } \alpha^5 = 0\}$. The proof uses only the $\mathbb{F}_q$ -translation and scaling symmetries of U .

- $d = 6$, **the constants (Sections 5–6)**. Theorem 5.3 reduces A_6, X, Y to explicit finite sums over $\mathbb{F}_{q^6}$. Theorem 6.1 expresses $A_6 - q^9$ and X through the Frobenius traces of the supersingular curves $y^2 + y = \lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x$, equivalently through $\mathbb{F}_2$ -quadratic forms on $\mathbb{F}_{q^{12}}$ with explicit radicals. Lemma 6.2 evaluates the pure cubic part for all k . Conjecture 6.4, verified exactly for $q \leq 128$ (Table 1), states $A_6 = q^9 + (-1)^{k+1}(q-1)q^5$ and $X = (-1)^{k+1}q^6 + q^5$.
- **Scaling law and Y (Section 7)**. Theorem 7.1 gives $\text{Var}_{\mathcal{G}^2}(\Psi)/q^7$ at $d = 6$ in closed form in terms of (q, k, m_k) , $m_k = Y/q^3$; it implies $\text{Var}_{\mathcal{G}^2}/q^7 \asymp q^2$ (Table 3) and $\lim \text{Var}_{\mathcal{G}^2}/q^9 = 1$ provided $m_k = o(q^{5/2})$. We identify Y as a Frobenius trace on the nine-dimensional variety U with an Artin–Schreier sheaf (Proposition 7.3) and show that the data $m_k = 13, -1, -35, 47, 733$ ($k \leq 5$) exclude a pure weight-3 object of rank ≤ 5 .

Methods. Two ideas carry the paper. The first is the *quadratic tower*: for $n = 2d$ the extension $\mathbb{F}_{q^n}/\mathbb{F}_{q^{n/2}}$ is quadratic, so every $\alpha \in \mathbb{F}_{q^n}$ is a root of $t^2 + at + n$ with $(a, n) \in \mathbb{F}_{q^{n/2}}^2$, and the elementary symmetric functions of the n conjugates of α become explicit polynomials in traces of (a, n) . Counting pairs (a, n) with the irreducibility condition $\text{Tr}_{q^{n/2}/2}(n/a^2) = 1$ turns the prime-counting problem into linear algebra over $\mathbb{F}_q$ and $\mathbb{F}_2$. The second is the use of the *symmetries of $U_d = \{p_j = 0, j \text{ odd} \leq \ell\}$* : U_d is stable under $\alpha \mapsto \alpha + c$ ($c \in \mathbb{F}_q$) and $\alpha \mapsto s\alpha$ ($s \in \mathbb{F}_q^\times$), and the elementary symmetric functions transform by binomial coefficients mod 2; this alone forces the shape of the Legendre profile.

Reproducibility. All statements labelled “verified” were checked with exact integer arithmetic; the scripts `legendre_var_check.py`, `legendre_d4_formula.py`, `legendre_q16_hayes.py`, `legendre_d6_tower.py` and `legendre_d6_qf.py` accompany the paper. The Hayes-group FFT of [2] was used as an independent check for $q \leq 32$ at $d = 6$ and $q \leq 128$ at $d = 4$.

2. PRELIMINARIES

2.1. Hayes group, fibres, Legendre intervals. Let $\mathcal{G} = \mathcal{G}_\ell = \{1 + c_1u + \cdots + c_\ell u^\ell\} \subset (\mathbb{F}_q[u]/u^{\ell+1})^\times$, $|\mathcal{G}| = q^\ell$. A monic $F = t^n + a_{n-1}t^{n-1} + \cdots + a_0$ maps to $F^*(u) = u^n F(1/u) \bmod u^{\ell+1}$, i.e. to its top ℓ coefficients $(c_1, \dots, c_\ell) = (a_{n-1}, \dots, a_{n-\ell})$; the map is multiplicative and each fibre has $q^{n-\ell} = q^{d+1}$ elements.

Lemma 2.1. *The Legendre interval $\mathcal{I}_f$ is the fibre over the class of f^2 , and the classes of squares form the subgroup $\mathcal{G}^2 = \{g^2\}$, of index q^m with $m = \#\{j \leq \ell : j \text{ odd}\} = \lfloor d/2 \rfloor$. Hence $\Psi(f) = \Psi(g)$ depends only on $g = [f^2] \in \mathcal{G}^2$.*

Proof. $f^2 + s$ has vanishing coefficients in all odd degrees $> d$, and $(1 + \sum c_j u^j)^2 = 1 + \sum c_j^2 u^{2j}$. The fibre of $[f^2]$ is $\{f^2 + s : \deg s \leq d\}$ since the top $\ell = d - 1$ coefficients of $f^2 + s$ are those of f^2 . $\square$

2.2. Ψ as a root count. For $\alpha \in \mathbb{F}_{q^n}$ write $\text{charpoly}(\alpha) = \prod_{i < n} (t - \alpha^{q^i})$. If $F = P^j$ with $\deg P = n/j$, exactly $n/j = \Lambda(F)$ elements α have $\text{charpoly}(\alpha) = F$. Hence

$$(1) \quad \Psi(g) = \sum_{F \in \text{fibre}(g)} \Lambda(F) = \#\{\alpha \in \mathbb{F}_{q^n} : (e_1, \dots, e_\ell)(\alpha) = g\},$$

where $e_j(\alpha)$ denotes the j -th elementary symmetric function of the n conjugates of α . In particular $\sum_g \Psi(g) = q^n$ and the mean over $\mathcal{G}$ is q^{d+1} .

2.3. Variances and quadratic twists. Put $\varphi = \Psi - q^{d+1}$ and

$$\text{Var}_{\mathcal{G}}(\Psi) = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \varphi(g)^2, \quad \text{Var}_{\mathcal{G}^2}(\Psi) = \frac{1}{|\mathcal{G}^2|} \sum_{g \in \mathcal{G}^2} \varphi(g)^2.$$

For a character χ of $\mathcal{G}$ let $\psi_n(\chi) = \sum_{\deg F=n} \Lambda(F) \chi(F)$. Then $\varphi(g) = |\mathcal{G}|^{-1} \sum_{\chi \neq 1} \psi_n(\chi) \bar{\chi}(g)$, and for ε a character trivial on $\mathcal{G}^2$,

$$(2) \quad \rho(\varepsilon) := \frac{\sum_{\chi} \psi_n(\chi) \overline{\psi_n(\chi \varepsilon)}}{\sum_{\chi \neq 1} |\psi_n(\chi)|^2} = \frac{\sum_g \varphi(g)^2 \varepsilon(g)}{\sum_g \varphi(g)^2}, \quad \frac{\text{Var}_{\mathcal{G}^2}}{\text{Var}_{\mathcal{G}}} = 1 + \sum_{\varepsilon \neq 1, \varepsilon|_{\mathcal{G}^2}=1} \rho(\varepsilon).$$

For $\chi \neq 1$ the L -function $L(u, \chi) = \sum_F \chi(F) u^{\deg F}$ is a polynomial of degree $\leq \ell - 1 = d - 2$ with inverse roots of absolute value $\sqrt{q}$, and $\psi_n(\chi) = -\sum_j \gamma_j^n$. Thus $\text{Var}_{\mathcal{G}}(\Psi)/q^{d+1} = |\mathcal{G}|^{-1} \sum_{\chi \neq 1} |\text{Tr } \Theta_\chi^n|^2$ with $\Theta_\chi \in U(d - 2)$ the normalised Frobenius; the Keating–Rudnick prediction is the Haar average $d - 2$.

2.4. Quadratic characters are Artin–Schreier characters.

Proposition 2.2. *Let $p_j(g)$ be the j -th power sum of the “roots” of $g \in \mathcal{G}_\ell$, defined by Newton’s identities (in characteristic 2: $p_j = \sum_{i < j} c_i p_{j-i} + [j \text{ odd}] c_j$). Then $p_j(gh) = p_j(g) + p_j(h)$, $p_{2j} = p_j^2$, and $g \mapsto (p_j(g))_{j \text{ odd} \leq \ell}$ induces an isomorphism $\mathcal{G}/\mathcal{G}^2 \cong \mathbb{F}_q^m$. The q^m quadratic characters are*

$$\varepsilon_\lambda(g) = (-1)^{\text{Tr}_{\mathbb{F}_q/\mathbb{F}_2} \sum_{j \text{ odd}} \lambda_j p_j(g)}, \quad \lambda \in \mathbb{F}_q^m,$$

and on a root α of a polynomial in the class g , $\varepsilon_\lambda(\alpha) = \psi_0(\sum_j \lambda_j \alpha^j)$ with ψ_0 the canonical additive character of $\mathbb{F}_{q^n}$. In particular ε_λ is primitive (does not factor through $\mathcal{G}_{\ell-1}$) if and only if $\lambda_\ell \neq 0$, which requires ℓ odd, i.e. d even.

Proof. Power sums are additive on products because roots are concatenated; $p_j(g^2) = 2p_j(g) = 0$ for odd j and $p_{2j} = p_j^2$ in characteristic 2. The map is onto $\mathbb{F}_q^m$ because $p_j = c_j + (\text{terms in } c_{<j})$ lets one solve for the odd c_j ; its kernel has order $q^{\ell-m} = |\mathcal{G}^2|$ and contains $\mathcal{G}^2$. Finally $p_j(\text{charpoly } \alpha) = \text{Tr}_{q^n/q}(\alpha^j)$ and $\text{Tr}_{q^n/2} \circ \text{Tr}_{q^n/q} = \text{Tr}_{q^n/2}$. $\square$

We write $\chi(x) = (-1)^{\text{Tr}_{q/2}(x)}$ for $x \in \mathbb{F}_q$ and $W = \{w \in \mathbb{F}_q : \text{Tr}_{q/2} w = 1\}$. Throughout, $\varepsilon_0 := (-1)^{k+1}$.

3. THE CASE $d = 4$: UNCONDITIONAL THEOREMS

Here $n = 8$, $\ell = 3$, $g = (c_1, c_2, c_3) = (a_7, a_6, a_5)$, $\mathcal{G}^2 = \{(0, c, 0)\}$ has order q , and $\mathcal{G}_0 := \{c_1 = 0\} \cong (\mathbb{F}_q^2, +)$ is a subgroup of order q^2 containing $\mathcal{G}^2$.

3.1. The quadratic tower.

Lemma 3.1. *For $\alpha \in \mathbb{F}_{q^8}$ put $a = \alpha + \alpha^{q^4}$, $n = \alpha^{1+q^4} \in \mathbb{F}_{q^4}$ and $\text{Tr} = \text{Tr}_{q^4/q}$. Then*

$$e_1(\alpha) = \text{Tr } a, \quad e_2(\alpha) = e_2(a) + \text{Tr } n, \quad e_3(\alpha) = e_3(a) + \text{Tr } a \text{ Tr } n + \text{Tr}(an),$$

where $e_j(a)$ are the elementary symmetric functions of the four conjugates of a . A pair (a, n) has exactly two preimages α if $t^2 + at + n$ is irreducible over $\mathbb{F}_{q^4}$ (equivalently $a \neq 0$ and $\text{Tr}_{q^4/2}(n/a^2) = 1$), one preimage if $a = 0$ ($\alpha \in \mathbb{F}_{q^4}$, $n = \alpha^2$), and none otherwise.

Proof. The eight conjugates of α are the roots of $\prod_{i < 4} (t^2 + a^{q^i} t + n^{q^i})$; expand. If $t^2 + at + n$ is reducible with $a \neq 0$ its roots $\beta \in \mathbb{F}_{q^4}$ satisfy $\beta + \beta^{q^4} = 0 \neq a$. $\square$

Two facts are used repeatedly: (i) $\text{Tr}_{q^8/2}$ vanishes on $\mathbb{F}_{q^4}$, $\text{Tr}_{q^4/q}$ vanishes on $\mathbb{F}_{q^2}$ and on $\mathbb{F}_q$; (ii) for $a \in \mathbb{F}_{q^4}$ the $\mathbb{F}_q$ -linear forms $\text{Tr}(n)$, $\text{Tr}(an)$ are independent iff $a \notin \mathbb{F}_q$, and the $\mathbb{F}_2$ -form $\text{Tr}_{q^4/2}(a^{-2}n)$ lies in the $\mathbb{F}_2$ -span of $\{\text{Tr}_{q^4/2}(cn) : c \in \mathbb{F}_q + \mathbb{F}_q a\}$ iff $a \in \mathbb{F}_{q^2}$.

3.2. The closed formula.

Theorem 3.2. *For every $q = 2^k$,*

$$\begin{aligned} \Psi(c_1, c_2, c_3) &= q^5 \quad (c_1 \neq 0), \\ \Psi(0, c_2, c_3) &= q^5 + q^3[c_3 = 0] + 2q^3 I(c_2, c_3) - 2q^2 K(c_2, c_3), \end{aligned}$$

where $I(c_2, c_3) = [c_2 c_3 \neq 0 \text{ and } \text{Tr}_{q/2}(c_2^3/c_3^2) = 1]$ and

$$K(c_2, c_3) = \sum_{w \in W} \sum_{v \in \mathbb{F}_q} \chi \left(\frac{c_2 v^2}{w} + \frac{\sqrt{c_2} v}{w} + \frac{c_3 v^3}{w^2} \right).$$

Proof. We count pairs (a, n) as in Lemma 3.1 with $(e_1, e_2, e_3) = (c_1, c_2, c_3)$, according to the degree of a over $\mathbb{F}_q$.

Case I: $a \notin \mathbb{F}_{q^2}$. We need $\text{Tr } a = c_1$ and then two independent $\mathbb{F}_q$ -linear conditions on n , $\text{Tr } n = c_2 + e_2(a)$ and $\text{Tr}(an) = c_3 + e_3(a) + c_1(c_2 + e_2(a))$, cutting out an affine subspace of size q^2 ; the irreducibility condition is a non-constant affine $\mathbb{F}_2$ -functional on it by fact (ii), so exactly $q^2/2$ values of n qualify, each with two roots. The number of a is $q^3 - q^2[c_1 = 0]$ (all $a \in \mathbb{F}_{q^2}$ have $\text{Tr } a = 0$). Contribution: $q^5 - q^4[c_1 = 0]$.

Case II: $a \in \mathbb{F}_{q^2} \setminus \mathbb{F}_q$. Then $\text{Tr } a = 0$, so $c_1 = 0$. Let $T = a + a^q \in \mathbb{F}_q^\times$, $N = a^{1+q}$; the conjugates are a, a^q, a, a^q , so $e_2(a) = T^2$ and $e_3(a) = 0$. The conditions are $\text{Tr } n = c_2 + T^2$, $\text{Tr}(an) = c_3$ (q^2 solutions). Writing $a^{-2} = x + ya$ with $x = (N + T^2)/N^2$, $y = T/N^2$ (from $a^2 = Ta + N$), the irreducibility functional equals $\text{Tr}_{q/2}(x(c_2 + T^2) + yc_3)$ on the solution set, and $\text{Tr}_{q/2}(xT^2) = \text{Tr}_{q/2}(z + z^2) = 0$ with $z = T^2/N$. Hence the contribution is $2q^2$ times the number of a with

$$\tau(a) := \text{Tr}_{q/2} \left(\frac{(N + T^2)c_2 + Tc_3}{N^2} \right) = 1.$$

Such a correspond two-to-one to irreducible $t^2 + Tt + N$, i.e. to (T, N) with $\text{Tr}_{q/2}(N/T^2) = 1$. Substituting $N = T^2w$ ($w \in W$), $v = 1/T$ and using $\text{Tr}_{q/2}(c_2v^2/w^2) = \text{Tr}_{q/2}(\sqrt{c_2}v/w)$ gives contribution $4q^2M$ with $M = \#\{(v, w) \in \mathbb{F}_q^\times \times W : \tau = 1\} = q^2/4 - K/2$, i.e. $q^4 - 2q^2K$.

Case III: $a \in \mathbb{F}_q^\times$. Then $c_1 = 0$, $e_2(a) = e_3(a) = 0$, the conditions read $\text{Tr } n = c_2$, $a \text{Tr } n = c_3$, and irreducibility is $\text{Tr}_{q/2}(a^{-2}c_2) = 1$. This forces $c_2 \neq 0$, $a = c_3/c_2 \neq 0$ and $\text{Tr}_{q/2}(c_2^3/c_3^2) = 1$; then n ranges over q^3 elements, each with two roots. Contribution $2q^3I(c_2, c_3)$.

Case IV: $a = 0$. $\alpha \in \mathbb{F}_{q^4}$, $(e_1, e_2, e_3) = (0, (\text{Tr } \alpha)^2, 0)$; contribution $q^3[c_1 = 0][c_3 = 0]$.

Adding the four contributions gives the theorem. $\square$

Corollary 3.3. *For every $q = 2^k$:*

- (a) $\varphi = \Psi - q^5$ is supported on $\mathcal{G}_0 = \{c_1 = 0\}$.
- (b) $\Psi(0, 0, 0) = \#\{\alpha \in \mathbb{F}_{q^8} : e_1 = e_2 = e_3 = 0\} = q^5 - q^4 + q^3$; equivalently there are exactly $q^4(q-1)/8$ irreducible octics with $a_7 = a_6 = a_5 = 0$.
- (c) For $c \neq 0$, $\Psi(0, c, 0) = q^5 + \varepsilon_0 q^3$, i.e. $q^5 + q^3$ for k even and $q^5 - q^3$ for k odd.
- (d) $A := \sum_c \Psi(0, c, 0) = \#\{\alpha : \text{Tr } \alpha = \text{Tr } \alpha^3 = 0\}$ equals q^6 for k even and $q^6 - 2q^4 + 2q^3$ for k odd.

Proof. (a) is the first line of the theorem. (b): $K(0, 0) = q^2/2$, $I(0, 0) = 0$. (c): with $y = \sqrt{c}v$ one has $\chi(cv^2/w + \sqrt{c}v/w) = \chi((y^2 + y)/w)$; since $y \mapsto y^2 + y$ is two-to-one onto $\{\text{Tr } z = 0\}$ and $z \mapsto \text{Tr}(z/w)$ is nontrivial there unless $w = 1$, $K(c, 0) = q[1 \in W] = q[k \text{ odd}]$. (d) follows from (b), (c). The prime-power part of $\Psi(0, c, 0)$ equals q^3 for every c (Part I), which gives the count of irreducibles in (b). $\square$

Theorem 3.4. *For every $q = 2^k$, $\frac{\text{Var}_{\mathcal{G}^2}(\Psi)}{q^5} = q(q-1)$.*

Proof. By Corollary 3.3(b), (c), $\sum_{g \in \mathcal{G}^2} \varphi(g)^2 = (q^4 - q^3)^2 + (q-1)q^6 = q^7(q-1)$, and $|\mathcal{G}^2| = q$. $\square$

Remark 3.5. The factor $q(q-1) = (q-1)^2 + (q-1)$ splits as the contribution of the identity class (the interval around t^8 , whose prime deficit is $q^4 - q^3$, relative size $1/q$) plus that of the other $q-1$ Legendre classes ($\pm q^3$ each). The sign in (c) depends on the parity of k but the variance does not.

3.3. Cubic Artin–Schreier sums over $\mathbb{F}_{q^8}$. Let $S(b, a) = \sum_{x \in \mathbb{F}_{q^8}} \psi_0(ax^3 + bx)$.

Lemma 3.6. *For $a \in \mathbb{F}_q^\times$ and $b \in \mathbb{F}_q$, $S(b, a) = S(0, a)$.*

Proof. For any $c \in \mathbb{F}_{q^8}$ the substitution $x \mapsto x + c$ gives $S(b, a) = \psi_0(ac^3 + bc)S(b + T_a(c), a)$ with $T_a(c) = ac^2 + \sqrt{ac}$, since $\text{Tr}(acx^2) = \text{Tr}(\sqrt{ac}x)$. If $c \in \mathbb{F}_{q^4}$ then $\text{Tr}_{q^8/2}(ac^3 + bc) = 0$. It remains to find $c \in \mathbb{F}_{q^4}$ with $T_a(c) = b$. The map T_a preserves $\mathbb{F}_{q^4}$ and is self-adjoint for $\langle u, v \rangle = \text{Tr}_{q^4/2}(uv)$, so $\text{im } T_a = (\ker T_a)^\perp$ with $\ker T_a = \{0\} \cup \{u \in \mathbb{F}_{q^4} : u^3 = a^{-1}\}$. For such $u \neq 0$, $\zeta = u^{q-1}$ is a cube root of unity. If $\zeta = 1$, $u \in \mathbb{F}_q$ and $\text{Tr}_{q^4/2}(bu) = \text{Tr}_{q^4/2}(4bu) = 0$. If $\zeta \neq 1$ and k is even, $\zeta^q = \zeta$ forces $u^{q^4} = \zeta^4 u \neq u$, impossible; if k is odd, $\zeta^q = \zeta^2$ gives $u^{q^2} = u$, so $u \in \mathbb{F}_{q^2}$ and $\text{Tr}_{q^4/2}(bu) = \text{Tr}_{q^2/2}(2bu) = 0$. Hence $b \perp \ker T_a$. $\square$

Proposition 3.7 (Carlitz's evaluation, elementary proof). *For $a \in \mathbb{F}_q^\times$ let $\nu = a$. Then*

$$S(0, a) = \sum_{x \in \mathbb{F}_{q^8}} \psi_0(ax^3) = \begin{cases} -2q^4, & a \text{ is a cube in } \mathbb{F}_{q^8}, \\ +q^4, & \text{otherwise,} \end{cases}$$

and $a \in \mathbb{F}_q^\times$ is a cube in $\mathbb{F}_{q^8}$ iff k is odd or a is a cube in $\mathbb{F}_q$.

Proof. By Lemma 3.6, $S(0, a) = q^{-1} \sum_{b \in \mathbb{F}_q} S(b, a) = \sum_{\alpha: \text{Tr } \alpha = 0} \psi_0(a\alpha^3) = \widehat{\Psi}(0, a)$ in the notation of (3) below, whose value is computed in Theorem 3.11. The cube criterion: $(q^8 - 1)/(q - 1) \equiv 8 \equiv 2 \pmod{3}$ when $q \equiv 1 \pmod{3}$, while $3 \mid (q^8 - 1)/(q - 1)$ when $q \equiv 2 \pmod{3}$. $\square$

Remark 3.8. The classical route [1] uses the cubic Gauss sum $g_4 = \sum_{y \in \mathbb{F}_4^\times} \chi_3(y)(-1)^{\text{Tr } y} = 2$ and the Hasse–Davenport relation $g_{4^m} = (-1)^{m-1} 2^m$, giving $\sum_{x \in \mathbb{F}_{2^{2m}}} \psi_0(ax^3) = (-1)^{m+1} 2^{m+1}$ for cubes and $(-1)^m 2^m$ otherwise; with $m = 4k$ this agrees with Proposition 3.7.

3.4. The full-group variance. Identify $\mathcal{G}_0 = \{(0, x, y)\}$ with $(\mathbb{F}_q^2, +)$ and set

$$(3) \quad \widehat{\Psi}(\mu, \nu) = \sum_{x, y \in \mathbb{F}_q} \Psi(0, x, y) \chi(\mu x + \nu y).$$

For $j \in \mathbb{F}_q^\times$ let $C_j(x) = x^3 + (1+j)x^2 + x + 1$, $R_j = \#\{x \in \mathbb{F}_q : C_j(x) = 0\}$, $L_j = \#\{x : C_j(x) = 0, \text{Tr}_{q/2} x = 1\}$ and $L'_j = R_j - L_j$.

Lemma 3.9. For $(\mu, \nu) \neq (0, 0)$:

$$\begin{aligned} \widehat{\Psi}(\mu, 0) &= -q^4 \quad (\mu \neq 0), & \widehat{\Psi}(0, \nu) &= q^4(1 - R_0(\nu) - 2[k \text{ odd}]) \quad (\nu \neq 0), \\ \widehat{\Psi}(\mu, \nu) &= q^4(1 - 3L_j - L'_j) & (\mu\nu \neq 0, j = \mu^3/\nu^2), \end{aligned}$$

where $R_0(\nu) = \#\{t : t^3 = \nu\}$ equals 1 for k odd and $3[\nu \text{ is a cube}]$ for k even. In particular $\widehat{\Psi}(0, \nu) = -2q^4$ for k odd, and for k even $-2q^4$ or $+q^4$ according as ν is a cube in $\mathbb{F}_q$ or not.

Proof. Transform the three terms of Theorem 3.2. The term $q^3[y = 0]$ gives $q^4[\mu = 0]$. For K , the identity $\sum_x \chi(x\alpha + \sqrt{x}\beta) = q[\sqrt{\alpha} = \beta]$ yields $\widehat{K}(\mu, \nu) = q^2 \#\{(v, w) \in \mathbb{F}_q \times W : v^3 = \nu w^2, v^2(w+1) = \mu w^2\}$. For $I = [x, y \neq 0] \frac{1-\chi(x^3/y^2)}{2}$, the substitution $x = yt$ gives $\sum_{x, y \neq 0} \chi(x^3/y^2 + \mu x + \nu y) = q R^*(\mu, \nu) - (q-1)$ with $R^* = \#\{t \neq 0 : t^3 + \mu t + \nu = 0\}$. The three cases follow; for $\mu\nu \neq 0$, dividing the two conditions in $\widehat{K}$ gives $v = \nu(w+1)/\mu$ and $\nu^2(w+1)^3 = \mu^3 w^2$, and $t = (\nu/\mu)s$, $s = 1 + 1/w$ transforms $t^3 + \mu t + \nu = 0$ into $C_j(w) = 0$, so $R^* = R_j$ and $\widehat{K} = q^2 L_j$. $\square$

Lemma 3.10. $\Sigma := \sum_{j \in \mathbb{F}_q^\times} (1 - 3L_j - L'_j)^2 = 3q - 4 + (-1)^k$.

Proof. Put $c(x) = 2 - \chi(x) \in \{1, 3\}$, so $3L_j + L'_j = \sum_{C_j(x)=0} c(x)$. Every $x \in \mathbb{F}_q \setminus \{0, 1\}$ is a root of exactly one C_j ($j = (x+1)^3/x^2$), and 0, 1 are roots of none. Expanding, $\Sigma = (q-1) - 2 \sum_{x \neq 0, 1} c(x) + \sum_{x \neq 0, 1} c(x)^2 + \text{OFF}$, where OFF is the sum of $c(x)c(x')$ over ordered pairs of *distinct* roots of a common C_j . Using $\sum_{x \neq 0, 1} \chi(x) = -1 - (-1)^k$ one finds $\Sigma = 2q - 1 + 2(-1)^k + \text{OFF}$. Distinct roots $x \neq x'$ of a common C_j are exactly the pairs with $x, x' \notin \{0, 1\}$ satisfying

$$(4) \quad x^2 x'^2 + x x' + x + x' = 0$$

(the third root is $1/(xx')$ and $e_2 = 1$; conversely (4) with $x, x' \notin \{0, 1\}$ produces a cubic C_j with $j \neq 0$). On (4) one has $x + x' = p^2 + p$ with $p = xx'$, hence $\chi(x)\chi(x') = 1$, and by symmetry $\text{OFF} = 5P - 4 \sum_{\text{pts}} \chi(x)$ where P is the number of points. For fixed $x \notin \{0, 1\}$, (4) is a quadratic in x' with $1 + \chi(x^3/(x+1)^2)$ roots, all $\notin \{0, 1\}$. With $y = x+1$: $x^3/(x+1)^2 = y+1+1/y+1/y^2$, whose χ -value is $(-1)^k \chi(y)$, and $x + x^3/(x+1)^2 = 1/y + 1/y^2$, whose χ -value is 1. Therefore $P = \sum_{\text{pts}} \chi(x) = q - 3 - (-1)^k$, $\text{OFF} = q - 3 - (-1)^k$ and $\Sigma = 3q - 4 + (-1)^k$. $\square$

Theorem 3.11. For every $q = 2^k$, $\frac{\text{Var}_{\mathcal{G}}(\Psi)}{q^5} = 3 - \frac{3}{q}$ and $\frac{\text{Var}_{\mathcal{G}^2}(\Psi)}{\text{Var}_{\mathcal{G}}(\Psi)} = \frac{q^2}{3}$.

Proof. By Corollary 3.3(a) and Parseval on $\mathcal{G}_0 \cong \mathbb{F}_q^2$ (note $\sum_{\mathcal{G}_0} \varphi = 0$),

$$\sum_{\mathcal{G}} \varphi^2 = \frac{1}{q^2} \sum_{(\mu, \nu) \neq (0,0)} |\widehat{\Psi}(\mu, \nu)|^2.$$

Lemma 3.9 gives $(q-1)q^8$ from $\nu = 0$, $\kappa_k(q-1)q^8$ from $\mu = 0$ with $\kappa_k = 4$ (k odd) and $\kappa_k = \frac{1}{3} \cdot 4 + \frac{2}{3} \cdot 1 = 2$ (k even), and $q^8(q-1)\Sigma$ from $\mu\nu \neq 0$ (each j has exactly $q-1$ preimages (μ, ν)). By Lemma 3.10 the bracket $1 + \kappa_k + \Sigma$ equals $3q$ in both parities, so $\sum_{\mathcal{G}} \varphi^2 = 3q^7(q-1)$. $\square$

Theorem 3.12. *Let $\varepsilon_{\lambda_1, \lambda_3}$ be the quadratic characters of $\mathcal{G}_3$ (Proposition 2.2). Then*

- (i) $\rho(\varepsilon_{\lambda_1, 0}) = 1$ for all $\lambda_1 \neq 0$ ($q-1$ characters of conductor u^2);
- (ii) $\rho(\varepsilon_{\lambda_1, \lambda_3}) = r(\lambda_3)$ is independent of λ_1 and constant on cosets of the cubes in $\mathbb{F}_q^\times$;
- (iii) if k is odd, $r \equiv \frac{q-3}{3(q-1)}$;
- (iv) $\rho_{\text{avg}} = \frac{q^2/3 - 1}{q^2 - 1} \rightarrow \frac{1}{3}$.

Proof. (i), (ii): φ is supported on $\mathcal{G}_0$ where $p_1 = 0$ and $p_3 = c_3$, so $\varepsilon_{\lambda_1, \lambda_3}|_{\mathcal{G}_0} = \chi(\lambda_3 c_3)$; the scaling automorphism $(c_1, c_2, c_3) \mapsto (sc_1, s^2 c_2, s^3 c_3)$ preserves Ψ . (iii): for k odd cubing is onto, so $\Phi(y) = \sum_x \varphi(0, x, y)^2$ is constant for $y \neq 0$, equal to $(3q^7(q-1) - q^7(q-1))/(q-1) = 2q^7$ by Theorems 3.4–3.11; then $r = (q^7(q-1) - 2q^7)/(3q^7(q-1))$. (iv) is immediate. $\square$

Remark 3.13. For $q = 8$ this gives seven characters with $\rho = 1$ and 56 with $\rho = 5/21 = 0.2381$, matching [2, Table 4] exactly; the “order-two coordinates” there contribute $7 \cdot 5/21 = 5/3$. The excess over Keating–Rudnick comes from the primitive quadratic characters, whose ρ does not decay in q ; $\sum_{\chi \neq 1} \psi_8(\chi) = -q^7 + q^6$ shows that $\mathbb{E}_\chi \text{Tr } \Theta_\chi^8 \rightarrow -1$, so the Frobenius classes are not Haar-distributed in $U(2)$ when $p = 2$, $\ell = 3$.

4. THE CASE $d = 6$: THE THREE-VALUE STRUCTURE THEOREM

Now $n = 12$, $\ell = 5$, $\mathcal{G}^2 = \{(0, c_2, 0, c_4, 0)\}$ has order q^2 and

$$U := \{\alpha \in \mathbb{F}_{q^{12}} : \text{Tr } \alpha = \text{Tr } \alpha^3 = \text{Tr } \alpha^5 = 0\}, \quad \text{Tr} = \text{Tr}_{q^{12}/q}.$$

Since $p_3 = c_3 + c_1 c_2 + c_1^3$ and $p_5 \equiv c_5$ modulo terms involving c_1, c_3 , the Legendre classes are exactly the classes with $p_1 = p_3 = p_5 = 0$, and $\Psi(0, c_2, 0, c_4, 0) = \#\{\alpha \in U : e_2(\alpha) = c_2, e_4(\alpha) = c_4\}$. Define

$$A_6 = |U|, \quad X = \sum_{\alpha \in U} \chi(e_2(\alpha)), \quad Y = \sum_{\alpha \in U} \chi(e_4(\alpha)).$$

Lemma 4.1. *U is stable under $\alpha \mapsto \alpha + c$ ($c \in \mathbb{F}_q$) and $\alpha \mapsto s\alpha$ ($s \in \mathbb{F}_q^\times$). On U ,*

$$e_2(\alpha + c) = e_2(\alpha), \quad e_4(\alpha + c) = e_4(\alpha) + c^2 e_2(\alpha) + c^4, \quad (e_2, e_4)(s\alpha) = (s^2 e_2, s^4 e_4)(\alpha).$$

Proof. $\text{Tr}(\alpha + c)^3 = \text{Tr } \alpha^3 + c(\text{Tr } \alpha)^2$ and $\text{Tr}(\alpha + c)^5 = \text{Tr } \alpha^5 + c(\text{Tr } \alpha)^4$ since $12c = 0$. For the elementary symmetric functions, $e_j(\alpha + c) = \sum_i \binom{12-i}{j-i} c^{j-i} e_i(\alpha)$; modulo 2, $\binom{12}{2} = 66$, $\binom{11}{1} = 11$, $\binom{12}{4} = 495$, $\binom{11}{3} = 165$, $\binom{10}{2} = 45$, $\binom{9}{1} = 9$, and on U one has $e_1 = e_3 = 0$. $\square$

Theorem 4.2 (three-value profile). *For all $q = 2^k$ and all $(c_2, c_4) \in \mathbb{F}_q^2$,*

$$\Psi(0, c_2, 0, c_4, 0) = q^7 + \frac{A_6 - q^9 - X}{q^2} + \frac{X}{q} [c_2 = 0] + \frac{Y}{q} [c_2 \neq 0] \chi\left(\frac{c_4}{c_2^2}\right).$$

Equivalently, the Fourier transform $\widehat{\Psi}_H(\mu, \nu) = \sum_{\alpha \in U} \chi(\mu e_2 + \nu e_4)$ equals $A_6, X, Y\chi(\mu/\sqrt{\nu})$ for $(\mu, \nu) = (0, 0)$, $(\mu \neq 0, 0)$, $\nu \neq 0$ respectively.

Proof. Let $F(\lambda) = \sum_U \chi(\lambda e_2 + e_4)$. By Lemma 4.1, $F(\lambda) = \sum_U \chi(\lambda e_2 + e_4 + c^2 e_2 + c^4) = \chi(c)F(\lambda + c^2)$, hence $F(\lambda) = \chi(\sqrt{\lambda})F(0) = \chi(\lambda)Y$. For $\nu \neq 0$ choose $s = \nu^{-1/4}$; scaling gives $\widehat{\Psi}_H(\mu, \nu) = F(\mu\nu^{-1/2}) = \chi(\mu/\sqrt{\nu})Y$, and for $\mu \neq 0$ scaling gives $\widehat{\Psi}_H(\mu, 0) = X$. Fourier inversion on $\mathbb{F}_q^2$ yields the displayed formula. $\square$

Corollary 4.3. Put $B = (A_6 - q^9 - X)/q^2$. Then $\varphi = \Psi - q^7$ equals $\varphi_0 = B + X/q$ on the q classes with $c_2 = 0$, $\varphi_A = B + Y/q$ on the $q(q-1)/2$ classes with $c_2 \neq 0$, $\text{Tr}_{q/2}(c_4/c_2^2) = 0$, and $\varphi_B = B - Y/q$ on the remaining $q(q-1)/2$ classes, and

$$(5) \quad \frac{\text{Var}_{\mathcal{G}^2}(\Psi)}{q^7} = \frac{q\varphi_0^2 + q(q-1)(B^2 + Y^2/q^2)}{q^9}.$$

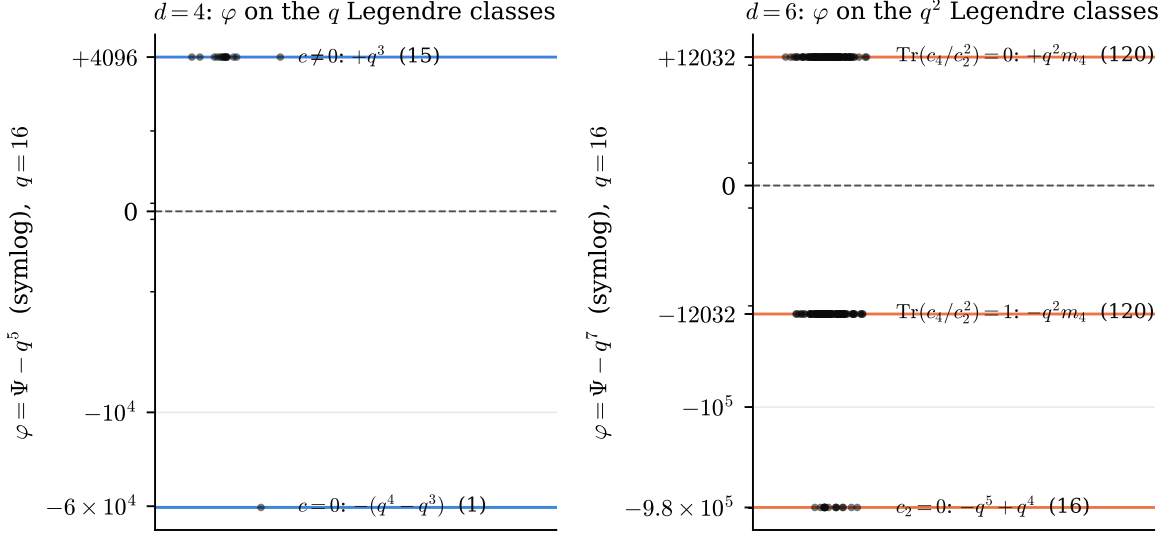


FIGURE 1. The Legendre deviations $\varphi = \Psi - q^{d+1}$ for $q = 16$. Left: $d = 4$, the q classes $(0, c, 0)$ take the two values of Corollary 3.3; the identity class carries $-(q^4 - q^3)$. Right: $d = 6$, the q^2 classes $(0, c_2, 0, c_4, 0)$ take exactly the three values of Theorem 4.2 ($m_4 = 47$). Each dot is one class (horizontal jitter only); the deviations are not continuous noise but are pinned to discrete levels by the translation symmetry of U .

Remark 4.4. The same argument applies to every d : $U_d = \{p_j = 0, j \text{ odd} \leq \ell\}$ is stable under $\mathbb{F}_q$ -translations and scalings, and the transformation law of e_{2j} is governed by $\binom{2d-i}{2j-i} \bmod 2$ (Lucas' theorem). For $d = 4$ one recovers Corollary 3.3(c).

5. THE QUADRATIC TOWER $\mathbb{F}_{q^{12}} \supset \mathbb{F}_{q^6}$

Lemma 5.1. For $\alpha \in \mathbb{F}_{q^{12}}$ let $a = \alpha + \alpha^{q^6}$, $n = \alpha^{1+q^6} \in \mathbb{F}_{q^6}$, $\text{Tr} = \text{Tr}_{q^6/q}$, and $e_j(a)$ the elementary symmetric functions of the six conjugates of a . Then

$$\begin{aligned} e_1 &= \text{Tr } a, & e_2 &= e_2(a) + \text{Tr } n, & e_3 &= e_3(a) + \text{Tr } a \text{Tr } n + \text{Tr}(an), \\ e_4 &= e_4(a) + e_2(a) \text{Tr } n + e_1(a) \text{Tr}(an) + \text{Tr}(a^2 n) + e_2(n), \\ e_5 &= e_5(a) + e_3(a) \text{Tr } n + e_2(a) \text{Tr}(an) + e_1(a) \text{Tr}(a^2 n) + \text{Tr}(a^3 n) \\ &\quad + e_1(a)e_2(n) + \text{Tr}(an) \text{Tr } n + \text{Tr}(an^2), \\ p_3 &= \text{Tr}(a^3 + an), & p_5 &= \text{Tr}(a^5 + a^3 n + an^2), \end{aligned}$$

and $e_2(\beta + \beta^q) = \text{Tr}_{q^{12}/q}(\beta^{q+1}) + (\text{Tr}_{q^{12}/q} \beta)^2$ for all $\beta \in \mathbb{F}_{q^{12}}$. The multiplicity statement of Lemma 3.1 holds verbatim with $\mathbb{F}_{q^4}$ replaced by $\mathbb{F}_{q^6}$; in particular $\mathbb{F}_{q^6} \subset U$, with $e_2|_{\mathbb{F}_{q^6}} = (\text{Tr } \alpha)^2$ and $e_4|_{\mathbb{F}_{q^6}} = e_2(\alpha)^2$.

Proof. Expand $\prod_{i < 6} (t^2 + a_i t + n_i)$ using $e_2(a \setminus j) = e_2(a) + a_j e_1(a) + a_j^2$, $e_3(a \setminus j) = e_3(a) + a_j e_2(a) + a_j^2 e_1(a) + a_j^3$ and $e_1(a \setminus j, k) = e_1(a) + a_j + a_k$. The power sums follow from $s_k = a s_{k-1} + n s_{k-2}$ for $s_k = \alpha^k + \alpha^{q^6 k}$. All identities were also checked by exhaustive computation over $\mathbb{F}_{2^{12}}$. $\square$

Lemma 5.2 (Gauss sum of e_2). *On $\mathbb{F}_{q^6}$ the quadratic form e_2 has polar form $B(x, y) = \text{Tr } x \text{Tr } y + \text{Tr}(xy)$, which is nondegenerate. For every $\beta \in \mathbb{F}_{q^6}$, $\sum_{n \in \mathbb{F}_{q^6}} \chi(e_2(n) + \text{Tr}(\beta n)) = \chi(e_2(\beta)) G$ with $G = \sum_n \chi(e_2(n))$.*

Proof. $n_0 = \beta + \text{Tr } \beta$ satisfies $B(n_0, \cdot) = \text{Tr}(\beta \cdot)$, and $e_2(\beta + c) = e_2(\beta) + c \text{Tr } \beta + c^2$ for $c \in \mathbb{F}_q$ gives $e_2(n_0) = e_2(\beta)$. $\square$

Theorem 5.3 (tower reduction). *Let a range over $\mathbb{F}_{q^6} \setminus \{0\}$ with $\text{Tr } a = 0$, $(\lambda, \mu) \in \mathbb{F}_q^2$, $\varepsilon \in \{0, 1\}$, and put*

$$\beta' = \lambda a + \sqrt{\mu} \sqrt{a} + \mu a^3 + \varepsilon a^{-2}, \quad \beta = e_2(a) + a^2 + \beta', \quad w = \chi(\lambda \text{Tr } a^3 + \mu \text{Tr } a^5).$$

Then

$$\begin{aligned} A_6 &= q^6 + q^4 \sum_a \sum_{\beta'=0} (-1)^\varepsilon w, & X &= q^4 \sum_a \chi(e_2(a)) \sum_{\beta'=1} (-1)^\varepsilon w, \\ Y &= G \left[1 + q^{-2} \sum_a \sum_{\lambda, \mu, \varepsilon} (-1)^\varepsilon \chi(e_4(a) + \lambda \text{Tr } a^3 + \mu \text{Tr } a^5 + e_2(\beta)) \right]. \end{aligned}$$

Proof. For $a \neq 0$ with $\text{Tr } a = 0$, the conditions $p_3 = p_5 = 0$ read $\text{Tr}(an) = \text{Tr } a^3$ and $\text{Tr}(\sqrt{a}n)^2 + \text{Tr}(a^3n) = \text{Tr } a^5$ (using $\text{Tr}(an^2) = \text{Tr}(\sqrt{a}n)^2$); expand their indicators and the irreducibility indicator $\frac{1}{2} \sum_\varepsilon (-1)^\varepsilon \chi(\varepsilon \text{Tr}(a^{-2}n))$ by additive characters, using $\chi(\mu t^2) = \chi(\sqrt{\mu}t)$. The n -sums are $q^6[\gamma = 0]$ for A_6, X (no $e_2(n)$ term) and $\chi(e_2(\beta))G$ for Y by Lemma 5.2. The contribution of $a \in \mathbb{F}_{q^6}$ is $q^6, 0, G$ respectively. $\square$

Remark 5.4. The equation $\beta' = 0$ with $\varepsilon = 0$ and $\mu \neq 0$ squares to $\mu^2 a^5 + \lambda^2 a + \mu = 0$, forcing $a \in \mathbb{F}_{q^2} \cup \mathbb{F}_{q^3}$ (a thin layer, as in Case II of Theorem 3.2). For $\varepsilon = 1$, with $u = \mu a^5 + \lambda a^3 + 1$ it becomes $u^2 + u = \lambda a^3 + 1$: the points $(a, u) \in E_\lambda(\mathbb{F}_{q^6})$ of the supersingular elliptic curve $E_\lambda : y^2 + y = \lambda x^3 + 1$ with $(u + \lambda a^3 + 1)/a^5 \in \mathbb{F}_q$. The additional term $e_2(\beta)$ in Y is quadratic in $(\lambda, \sqrt{\mu})$ with cubic cross terms, which is why Y does not collapse onto a thin layer.

The reductions were verified numerically for $q = 2, 4, 8$; they give $G = \varepsilon_0 q^3$ for $k \leq 3$.

6. QUADRATIC FORMS AND SUPERSINGULAR ARTIN-SCHREIER CURVES

6.1. A_6 and X as sums of Frobenius traces.

Theorem 6.1. *Write $\lambda = (\lambda_1, \lambda_3, \lambda_5) \in \mathbb{F}_q^3$ and $S(\lambda) = \sum_{x \in \mathbb{F}_{q^{12}}} \psi_0(\lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x)$. Then*

$$(6) \quad A_6 - q^9 = \frac{1}{q^3} \sum_{\lambda \neq 0} S(\lambda),$$

and $S(\lambda) = -\sum_j \omega_j(\lambda)^{12}$, where $\omega_j(\lambda)$ are the Frobenius eigenvalues over $\mathbb{F}_q$ of the Artin-Schreier curve $C_\lambda : y^2 + y = \lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x$ (genus 2, 1, 0 according as $\lambda_5 \neq 0$; $\lambda_5 = 0 \neq \lambda_3$; only $\lambda_1 \neq 0$). Moreover $\text{Tr}_{q^{12}/2}(\lambda_5 x^5 + \lambda_3 x^3) = Q_\lambda(x)$ is an $\mathbb{F}_2$ -quadratic form with polar form $B_\lambda(x, y) = \text{Tr}(y T_\lambda(x))$,

$$T_\lambda(x) = \lambda_5 x^4 + (\lambda_5 x)^{1/4} + \lambda_3 x^2 + (\lambda_3 x)^{1/2}, \quad R_\lambda = \ker T_\lambda = \{x : \lambda_5^4 x^{16} + \lambda_3^4 x^8 + \lambda_3^2 x^2 + \lambda_5 x = 0\},$$

so that $S(\lambda) \in \{0, \pm 2^{(12k+r_\lambda)/2}\}$ with $r_\lambda = \dim_{\mathbb{F}_2} R_\lambda \leq 4$, $S(\lambda) = 0$ iff $\text{Tr}(\lambda_1 x) \neq Q_\lambda(x)$ for some $x \in R_\lambda$, and the sign otherwise is the Arf invariant of Q_λ on $\mathbb{F}_{q^{12}}/R_\lambda$ (shifted by $Q_\lambda(x_0)$, $B_\lambda(x_0, \cdot) = \text{Tr}(\lambda_1 \cdot)$). In particular all C_λ are supersingular: $\omega_j(\lambda) = \sqrt{q} \zeta_j$ with roots of unity ζ_j .

Proof. $A_6 = \sum_{g \in \mathcal{G}^2} \Psi(g) = \frac{|\mathcal{G}^2|}{|\mathcal{G}|} \sum_{\varepsilon^2=1} \psi_{12}(\varepsilon)$ and $\psi_{12}(\varepsilon_\lambda) = S(\lambda)$ by Proposition 2.2; $S(0) = q^{12}$. The L -function interpretation is standard. $Q_\lambda(x) = \text{Tr}(x(\lambda_5 x^4 + \lambda_3 x^2))$ is of the form $\text{Tr}(xL(x))$ with L linearised, whence the quadratic-form statements; the kernel of T_λ equals that of T_λ^4 . Since the same structure persists over every $\mathbb{F}_{q^m}$, $\sum_{x \in \mathbb{F}_{q^m}} (-1)^{\text{Tr } f_\lambda(x)} \in \{0, \pm 2^{(mk+r)/2}\}$ for all m , which forces the normalised eigenvalues to be roots of unity [5]. $\square$

Summing over λ_1 and using Hilbert 90 ($\alpha = \beta + \beta^q$, $\theta = \sigma + \sigma^{-1}$, $\text{Tr } \alpha^3 = \text{Tr}(\beta^2\theta\beta)$, $\text{Tr } \alpha^5 = \text{Tr}(\beta^4\theta\beta)$) one obtains the equivalent forms

$$(7) \quad A_6 = q^9 + \frac{1}{q^3} \sum_{(\lambda_3, \lambda_5) \neq 0} W(\lambda_3, \lambda_5), \quad X = \frac{1}{q^3} \sum_{\lambda_3, \lambda_5} W'(\lambda_3, \lambda_5),$$

with $W = \sum_{\beta} (-1)^{\text{Tr}(\lambda_3 \beta^2 \theta \beta + \lambda_5 \beta^4 \theta \beta)}$ and $W' = \sum_{\beta} (-1)^{\text{Tr}(\lambda_3 \beta^2 \theta \beta + \lambda_5 \beta^4 \theta \beta + \beta^{q+1} + \beta)}$ (the second uses Lemma 5.1). Their polar forms are $\text{Tr}(\gamma M(\theta\beta))$ with $M(y) = \lambda_3 y^2 + \sqrt{\lambda_3 y} + \lambda_5 y^4 + (\lambda_5 y)^{1/4}$ (+ y for W'), so the radicals are $\theta^{-1}(\ker M \cap \text{im } \theta)$, of dimension $2k + \dim(\ker M \cap \text{im } \theta)$, where $\ker \theta = \mathbb{F}_{q^2}$ and $\text{im } \theta = \ker \text{Tr}_{q^{12}/q^2}$.

6.2. The pure cubic part.

Lemma 6.2. *For all k and all $\lambda_3 \in \mathbb{F}_q^\times$: $W(\lambda_3, 0) = 0$ if k is odd and $W(\lambda_3, 0) = -2q^7$ if k is even.*

Proof. $\ker M = \{0\} \cup \{y : y^3 = \lambda_3^{-1}\}$, and all three cube roots lie in $\mathbb{F}_{q^{12}}$. With $\zeta = y^{q-1} \in \mu_3$: $\zeta = 1$ gives $y \in \mathbb{F}_q$; for $\zeta \neq 1$, k odd gives $y \in \mathbb{F}_{q^2}$ and k even gives $y \in \mathbb{F}_{q^3}$. In all cases $\text{Tr}_{q^{12}/q^2} y = 0$, so $\ker M \subset \text{im } \theta$ and $\dim R = 2k + 2$, $|W| \in \{0, 2^{7k+1}\}$. Now $Q(\beta) = \text{Tr}_{q^{12}/2}(\lambda_3 \beta^2 \theta \beta)$ vanishes on $\ker \theta$, and for $\theta\beta = y$ it is independent of the choice of β (the ambiguity $c \in \mathbb{F}_{q^2}$ contributes $\text{Tr}(\lambda_3 c^2 y)$ with $c^2 y$ in a subfield of even index). If $y \in \mathbb{F}_{q^2}$, $\zeta \neq 1$: on $\mathbb{F}_{q^4}$ one has $\theta\beta = \sigma(\text{Tr}_{q^4/q^2} \beta)$, so pick $\beta_y \in \mathbb{F}_{q^4}$ with $\text{Tr}_{q^4/q^2} \beta_y = y^q$; then $Q(\beta_y) = \text{Tr}_{q^4/2}(\lambda_3 y \beta_y^2) = \text{Tr}_{q^2/2}(\lambda_3 y (\text{Tr}_{q^4/q^2} \beta_y)^2) = \text{Tr}_{q^2/2}(\lambda_3 y^{2q+1}) = \text{Tr}_{q^2/2}(\zeta^2) = k \pmod{2}$, which is 1 for k odd: $Q|_R \neq 0$ and $W = 0$. If $y \in \mathbb{F}_q$ the same computation gives $\text{Tr}_{q^2/2}(1) = 0$; if $y \in \mathbb{F}_{q^3}$ (k even) then $\text{Tr}_{q^3/q} y = 0$, θ is invertible on the trace-zero part of $\mathbb{F}_{q^3}$, $\beta_y \in \mathbb{F}_{q^3}$ and $Q(\beta_y) = 0$. So for k even $|W| = 2q^7$. Finally $W = \sum_{\lambda_1 \in \mathbb{F}_q} S(\lambda_1, \lambda_3, 0)$ with $|S| = 2q^6$ for each term ($r = 2$), and $|W| = q \cdot 2q^6$ forces all q signs to agree with that of $S(0, \lambda_3, 0) = -2q^6$ (Hasse–Davenport, $m = 6k$). $\square$

Remark 6.3. For k odd the analogue of Lemma 3.6 fails over $\mathbb{F}_{q^{12}}$: the cube roots lie in $\mathbb{F}_{q^2}$ and $[\mathbb{F}_{q^6} : \mathbb{F}_{q^2}] = 3$ is odd, so $\mathbb{F}_q \not\subset T(\mathbb{F}_{q^6})$. The vanishing of $W(\lambda_3, 0)$ is exactly this failure.

6.3. Exact evaluation and the conjecture. Every W, W' is the exponential sum of an $\mathbb{F}_2$ -quadratic form (plus a linear form) on $\mathbb{F}_2^{12k}$, hence computable exactly by linear algebra: radical, test of $Q + L$ on it, a symplectic basis of a complement and its Arf invariant. We implemented this (`legendre_d6_qf.py`) with $\mathbb{F}_{q^{12}} = \mathbb{F}_q[z]/(m)$ and the maps $\beta \mapsto \beta^2, \beta^{1/2}, \sigma, \theta$ as $12k \times 12k$ matrices over $\mathbb{F}_2$.

Conjecture 6.4. *For all $q = 2^k$: $A_6 = q^9 + \varepsilon_0(q-1)q^5$ and $X = \varepsilon_0 q^6 + q^5$, i.e.*

$$\sum_{\lambda_5 \neq 0} W(0, \lambda_5) + \sum_{\lambda_3 \lambda_5 \neq 0} W(\lambda_3, \lambda_5) = (q-1)q^7(\varepsilon_0 q + 2), \quad \sum_{\lambda_3, \lambda_5} W'(\lambda_3, \lambda_5) = q^3(\varepsilon_0 q^6 + q^5).$$

Proposition 6.5 (structure of the data). *For $k \leq 7$ the following hold.*

- (a) k odd: $W(0, \lambda_5) = 2q^7$ for all $\lambda_5 \neq 0$; the mixed W depends only on $j = \lambda_3^5/\lambda_5^3 \in \mathbb{F}_q^\times$ and equals $2q^7$ for exactly $(q-2)/2$ values of j and 0 otherwise. Hence $\sum W = (q-1)2q^7 + \frac{(q-1)(q-2)}{2} 2q^7 = (q-1)q^8$.
- (b) k even: $W(0, \lambda_5) = 0$ if $k \equiv 2 \pmod{4}$, while for $k \equiv 0 \pmod{4}$ (so $5 \mid q-1$) it equals $-4q^7$ for the $(q-1)/5$ fifth powers and $+q^7$ otherwise; the mixed W take values $-4q^7, q^7, 0$ with multiplicities $(q-1)(n_-, n_+, n_0)$, $(n_-, n_+, n_0) = (1, 2, 0), (5, 6, 4), (22, 26, 15)$ for $k = 2, 4, 6$, satisfying $n_+ - 4n_- = 2 - q$.

The set of j in (a) is not of the form $\{\text{Tr}_{q/2}(j^m) = c\}$ for any m when $k = 5, 7$ (it is $\{\text{Tr } j = 0\}$ when $k = 3$).

Conjecture 6.4 is thus equivalent to a precise statement about which radicals $\theta^{-1}(\ker M \cap \text{im } \theta)$ carry a vanishing Q and about the Arf invariants, for the 15-th degree linearised polynomials $\lambda_5^4 y^{15} + \lambda_3^4 y^7 + \lambda_3^2 y + \lambda_5$. The supersingular point counts of Theorem 6.1 determine the possible

TABLE 1. Exact values of A_6 and X by the quadratic-form algorithm ($N = 12k$ bits). For $k \leq 5$ the values coincide with the Hayes-group FFT of [2]; $k = 6, 7$ are beyond its reach at $d = 6$. In all cases $A_6 = q^9 + (-1)^{k+1}(q-1)q^5$ and $X = (-1)^{k+1}q^6 + q^5$.

k	q	N	A_6	X	time
1	2	12	544	96	< 1 s
2	4	24	259 072	-3 072	< 1 s
3	8	36	134 447 104	294 912	< 1 s
4	16	48	68 703 748 096	-15 728 640	2 s
5	32	60	35 185 412 276 224	1 107 296 256	9 s
6	64	72	18 014 330 863 747 072	-67 645 734 912	48 s
7	128	84	9 223 376 400 541 548 544	4 432 406 249 472	225 s

TABLE 2. Distribution of $W(\lambda_3, \lambda_5)/q^7$ over the $q^2 - 1$ nontrivial pairs (radical dimension in parentheses); “pure cubic” $\lambda_5 = 0$, “pure quintic” $\lambda_3 = 0$, “mixed” $\lambda_3\lambda_5 \neq 0$.

k	pure cubic ($q-1$)	pure quintic ($q-1$)	mixed $((q-1)^2)$
1	0 (4)	2 (4)	0 (6) $\times 1$
3	0 (8)	2 (8)	2 (8) $\times 21$; 0 (10) $\times 28$
5	0 (12)	2 (12)	2 (12) $\times 465$; 0 (14) $\times 496$
7	0 (16)	2 (16)	2 (16) $\times 8001$; 0 (18) $\times 8128$
2	-2 (6)	0 (8)	-4 (8) $\times 3$; 1 (4) $\times 6$
4	-2 (10)	-4 (12) $\times 3$; 1 (8) $\times 12$	-4 (12) $\times 75$; 1 (8) $\times 90$; 0 (12) $\times 60$
6	-2 (14)	0 (16)	-4 (16) $\times 1386$; 1 (12) $\times 1638$; 0 (16) $\times 945$

absolute values; they do not by themselves decide which pairs (λ_3, λ_5) vanish. We leave the pure quintic and mixed cases open.

7. THE SCALING LAW AT $d = 6$ AND THE NATURE OF Y

7.1. Variance formula.

Theorem 7.1. Assume Conjecture 6.4 (true for $q \leq 128$) and write $Y = q^3 m_k$. Then $B = -2q^3[k \text{ odd}]$,

$$\varphi_0 = \begin{cases} q^5 + q^4 - 2q^3, & k \text{ odd}, \\ -q^5 + q^4, & k \text{ even}, \end{cases} \quad \varphi_{A,B} = \begin{cases} -2q^3 \pm q^2 m_k, & k \text{ odd}, \\ \pm q^2 m_k, & k \text{ even}, \end{cases}$$

and

$$\frac{\text{Var}_{\mathcal{G}^2}(\Psi)}{q^7} = \begin{cases} \frac{(q-1)^2(q+2)^2}{q^2} + \frac{(q-1)(4q^2 + m_k^2)}{q^4}, & k \text{ odd}, \\ (q-1)^2 + \frac{(q-1)m_k^2}{q^4}, & k \text{ even}. \end{cases}$$

In particular $\text{Var}_{\mathcal{G}^2}(\Psi)/q^7 = q^2(1 + O(1/q))$, i.e. $C_6 := \lim_{q \rightarrow \infty} \text{Var}_{\mathcal{G}^2}(\Psi)/q^9 = 1$, provided $m_k = o(q^{5/2})$.

Proof. Substitute into Corollary 4.3. □

Remark 7.2. At $d = 4$ the Legendre anomaly is a single class with relative deficit $1/q$; at $d = 6$ it is the q classes with $c_2 = 0$ (the intervals around f^2 with $f_5 = 0$), each with relative deviation $\pm q^{-2}$ and sign ε_0 . Both mechanisms give $\text{Var}_{\mathcal{G}^2}/q^{d+1} \asymp q^2$. The full-group variance $\text{Var}_{\mathcal{G}}/q^7$

TABLE 3. $d = 6$: exact variances (Hayes FFT, $q \leq 32$; $q = 16$ also by CRT), the three Legendre deviations and $m_k = Y/q^3$. The last two columns show that $\text{Var}_{\mathcal{G}^2}/q^7 \asymp q^2$ and not q^4 .

q	$\text{Var}_{\mathcal{G}}/q^7$	$\text{Var}_{\mathcal{G}^2}/q^7$	ratio	φ_0	φ_A	φ_B	m_k	$\text{Var}_{\mathcal{G}^2}/q^9$
2	5.7578	15.5625	2.703	32	36	-68	13	3.891
4	3.7004	9.0117	2.435	-768	-16	16	-1	0.563
8	4.7483	79.0935	16.657	35 840	-3 264	1 216	-35	1.236
16	4.3514	225.5056	51.824	-983 040	12 032	-12 032	47	0.881
32	4.2086	1100.8844	261.581	34 537 472	685 056	-816 128	733	1.075

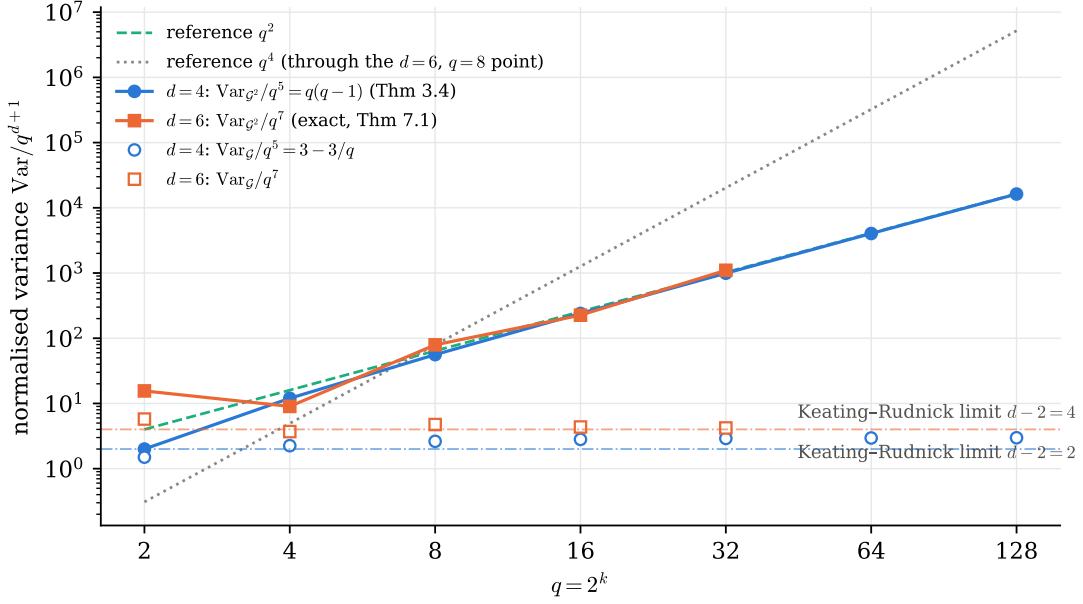


FIGURE 2. Normalised variances against q (log-log). Filled markers: $\text{Var}_{\mathcal{G}^2}(\Psi)/q^{d+1}$, exact values ($d = 4$: $q(q-1)$ for $q \leq 128$, Theorem 3.4; $d = 6$: Table 3). Open markers: the full-group variances $\text{Var}_{\mathcal{G}}/q^{d+1}$, which approach 3 ($d = 4$, Theorem 3.11) and the Keating–Rudnick value 4 ($d = 6$). The dashed line is q^2 ; the dotted line is a q^4 law normalised through the $d = 6$, $q = 8$ point, which the data exclude. Whether $\lim \text{Var}_{\mathcal{G}^2}/q^9 = 1$ at $d = 6$ depends on $m_k = o(q^{5/2})$ (Theorem 7.1).

tends to the Keating–Rudnick value 4 (4.35, 4.21 at $q = 16, 32$), unlike $d = 4$ where it tends to $3 \neq 2$. The value $q^{2(d/2-1)} = q^4$ suggested in [2] is excluded.

7.2. Y is a Frobenius trace, not a character sum.

Proposition 7.3. *Identify $\mathbb{F}_{q^{12}}$ with $\mathbb{F}_q^{12}$. The maps p_j and e_j are polynomial maps $\mathbb{A}^{12} \rightarrow \mathbb{A}^1$ over $\mathbb{F}_q$ of degree j , and $V = \{p_1 = p_3 = p_5 = 0\} \subset \mathbb{A}_{\mathbb{F}_q}^{12}$ is a variety of dimension 9 with $V(\mathbb{F}_q) = U$. Then*

$$Y = \sum_{P \in V(\mathbb{F}_q)} \psi_q(e_4(P)) = \sum_i (-1)^i \text{Tr}(\text{Frob}_q \mid H_c^i(V_{\mathbb{F}_q}, e_4^* \mathcal{L}_\psi)),$$

and by Deligne [4] the eigenvalues of Frob_q on H_c^i have absolute values $\leq q^{i/2}$.

By contrast with A_6 and X , the function $(-1)^{\text{Tr } e_4}$ on $\mathcal{G}^2$ is not a character: in Witt coordinates $\mathcal{G}^2 \cong W_2(\mathbb{F}_q)$, $(x_1, x_2) + (x'_1, x'_2) = (x_1 + x'_1, x_2 + x'_2 + x_1 x'_1)$, and the carry destroys multiplicativity

(the characters of order 4 are $i^{\text{Tr } x_1}(-1)^{\text{Tr } x_2 + \dots}$). Therefore Y does not decompose into Frobenius traces of individual L -functions and no supersingularity argument applies.

Proposition 7.4. *The values $m_k = 13, -1, -35, 47, 733$ ($k = 1, \dots, 5$) satisfy $|m_k|/q^{3/2} = 4.60, 0.13, 1.55, 0.73, 4.05$. Consequently, if $Y_k = \pm \sum_{j=1}^b \omega_j^k$ with $|\omega_j| = 2^{9/2}$ (pure weight 9, rank b), then $b \geq 5$. The elementary symmetric functions of the would-be $\eta_j = \omega_j/8$ computed from $m_1, \dots, m_5$ by Newton's identities are 13, 85, 361, 1069, 2281, all integers; since the product of five algebraic integers of absolute value $2^{3/2}$ is not rational, pure weight 3 with $b = 5$ is impossible. The data are consistent with even rank $b \geq 6$, or with a dominant weight-9 part together with lower-weight contributions.*

Remark 7.5 (what is and is not established). Proposition 7.3 is the trace formula and is unconditional. We have *not* shown that the cohomology is concentrated in degree 9 (which would require an analysis of $e_4|_V$ at infinity in the spirit of Katz–Laumon), nor computed the rank. Hence the bound $m_k = O(q^{3/2})$, and with it $C_6 = 1$, is supported by the data but not proved. Note also that V depends on q (it is cut out of $\text{Res}_{\mathbb{F}_{q^{12}}/\mathbb{F}_q} \mathbb{A}^1$ by relative traces); the natural $\mathbb{F}_2$ -rational object is the moduli description $\Psi(g) = \#\{(F, \alpha) : F \in \text{fibre}(g), F(\alpha) = 0, \alpha \in \mathbb{F}_{q^{12}}\}$ over the parameter space $\mathbb{A}^9 = \{(c_2, c_4, \dots, c_{12})\}$, and $(Y_k)_k$ should be the $\mathbb{F}_{2^k}$ -traces of an ℓ -adic complex on it, whose L -function is the genuine “closed form” of Y . The tower reduction of Theorem 5.3 exhibits $Y/G - 1$ as a sum over $a \in \mathbb{F}_{q^6}$ of cubic Artin–Schreier sums over $\mathbb{F}_q$, i.e. as the trace of a relative H^1 of an elliptic Artin–Schreier family over a five-dimensional base; this is the natural starting point for identifying the complex.

8. SUMMARY OF THE NUMERICAL EVIDENCE

TABLE 4. Status of the main identities. “exact” means verified by exact integer computation; the method is indicated.

Statement	Status	Range checked
$d = 4$: Theorem 3.2 (all classes)	proved	formula = FFT class by class, $q \leq 128$
$d = 4$: $\text{Var}_{\mathcal{G}^2}/q^5 = q(q-1)$, $\text{Var}_{\mathcal{G}}/q^5 = 3 - 3/q$	proved	Lemma 3.10 also checked to $q = 2^{22}$
$d = 4$: ρ structure (Theorem 3.12)	proved	matches [2, Table 4]
$d = 6$: three-value profile (Theorem 4.2)	proved	all q^2 classes, $q \leq 32$
$d = 6$: tower identities and reductions	proved	$\mathbb{F}_{2^{12}}$ exhaustive; $q = 2, 4, 8$
$d = 6$: A_6, X closed forms (Conjecture 6.4)	open; pure cubic part proved	exact, $q \leq 128$ (Table 1)
$d = 6$: $\text{Var}_{\mathcal{G}^2}/q^7$ formula (Theorem 7.1)	proved given Conj. 6.4	$q \leq 32$
$C_6 = 1$	conditional on $m_k = o(q^{5/2})$	$q \leq 32$

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GUT GEOSERVICE INC., MONTREAL, CANADA
Email address: ruqing@hotmail.com